Python Assignment 2
Data Structures
And
Conditional
Statements

Objective

To understand and implement Python data structures such as lists, dictionaries, and sets, and to apply conditional statements using if, elif, and else.

Topics Covered

1. **List Operations:** Creating lists, appending, inserting, removing, popping, extending, sorting, accessing elements, and finding maximum, minimum, and sum values.
2. **Dictionary Operations:** Creating a dictionary, accessing student marks, adding and updating entries, and using keys(), values(), and items() methods.
3. **Set Operations:** Creating sets, understanding duplicate removal, identifying set indexing errors, and performing union and intersection operations.
4. **Conditional Statements:** Developing a performance category program using if, elif, and else based on a score between 0 and 10.

Conclusion

This assignment provided practical knowledge of Python data structures and their operations. It helped develop an understanding of list manipulation, dictionary management, set properties, and conditional statements. The performance category program demonstrated how to validate user input and classify scores based on specified conditions.